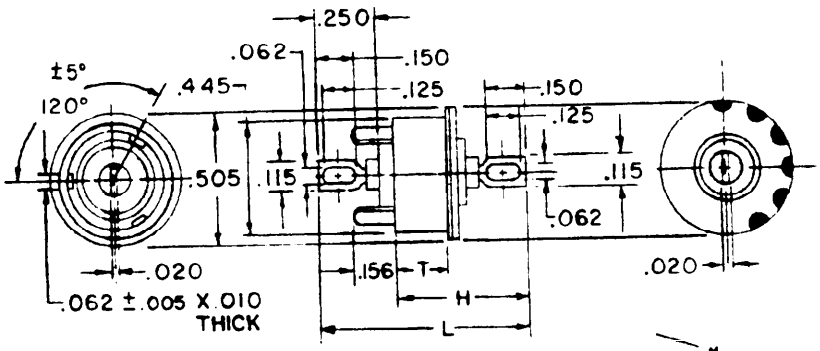
MILITARY SPECIFICATION SHEET

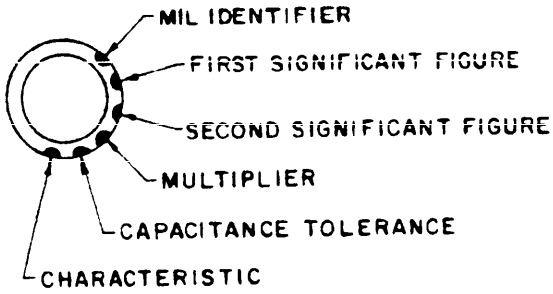
CAPACITORS, FIXED, MICA DIELECTRIC, BUTTON STYLE (FEEDTHRU, HERMETIC SEAL) STYLES CB55, CB56, AND CB57

This specification is approved for use by all Departments and Agencies of the Department of Defense.

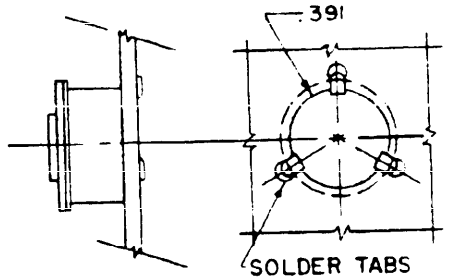
The complete requirements for procuring the capacitors described herein shall consist of this document and the latest issue of Specification MIL-C-10950.



INCHES	MM
.005	.13
.010	.25
.020	.51
.062	1.57
.115	2.92
.125	3.18
.150	3.81
.156	3.96
.250	6.35
.391	9.93
.445	11.30
.505	12.83



COLOR CODE MARKING (SEE NOTE 4)



MOUNTING SKETCH

Style	Dimensions T±.015(.38)	L±.045(1.14)	H±.030(.76)
CB55	.100 (2.54)	.710 (18.03)	.415 (10.54)
CB56	.125 (3.18)	.735 (18.67)	.440 (11.18)
CB57	.205 (5.21)	.815 (20.70)	.515 (13.08)



CIRCUIT DIAGRAM

NOTES:
1. All dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Unless otherwise specified, tolerance is .xxx ±.010 (.25 mm).
4. Color-coded section shall not exceed 240°.
5. Ferrous material may be used.
6. It is recommended that capacitor body have adequate heat sink during mounting operation with high temperature solder.

FIGURE 1. Capacitors, (feedthru, hermetic seal), styles CB55, CB56, and CB57.

Ⓒ denotes changes

FSC 5910

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TABLE I. Styles CB55, CB56, and CB57.

Part number 1/	Capacitance	DC rated voltage	Characteristic available at capacitance tolerance				Terminal assemblies
			+1% (F)	+2% (G)	+5% (J)	+10% (K)	
	pF	Volts					
CB55RB050K	5	500				B	R
CB55R-100K	10	500				D	R
CB55R-150K	15	500				D	R
CB55R-200J	20	500			D	-	R
CB55R-220-	22	500			D	D	R
CB55R-330-	33	500			D	D	R
CB55R-470-	47	500			D	D	R
CB55R-680-	68	500	D,F	D,F	D,F	D	R
CB55R-101-	100	500	D,F	D,F	D,F	D	R
CB55R-151-	150	500	D,F	D,F	D,F	D	R
CB55R-221-	220	500	D,F	D,F	D,F	D	R
CB55R-331-	330	500	D,F	D,F	D,F	D	R
CB55R-471-	470	500	D,F	D,F	D,F	D	R
CB56R-681-	680	500	D,F	D,F	D,F	D	R
CB56R-821-	820	500	D,F	D,F	D,F	D	R
CB56R-102-	1,000	500	D,F	D,F	D,F	D	R
CB57R-122-	1,200	500	D,F	D,F	D,F	D	R
CB57R-152-	1,500	500	D,F	D,F	D,F	D	R
CB57R-182-	1,800	500	D,F	D,F	D,F	D	R
CB57R-222-	2,200	500	D,F	D,F	D,F	D	R
CB57R-242-	2,400	500	D,F	D,F	D,F	D	R

1/ Complete part number will include additional symbols to indicate characteristic and capacitance tolerance.

REQUIREMENTS:

Design and construction:
Dimensions and configuration: See figure 1.
Case: See figure 1.
Terminals: See figure 1.
Capacitance value: See table I.
Capacitance tolerance: See table I.
DC rated voltage: 500 volts.
Dissipation factor: See figure 1 of MIL-C-10950.
Characteristics: See table I.

Symbol	Temperature coefficient	Capacitance drift
B	Not specified	Not specified
D	-100 to +100	0.3 percent or 0.3 pF whichever is greater.
F	-60 to +60	±(0.1 percent + 0.1 pF).

Operating temperature range: -55° to +150°C.

Dielectric withstanding voltage: Method 301 of MIL-STD-202.
200 percent of rated dc voltage for not less than 1 second, nor more than 5 seconds.

Insulation resistance: Method 302 of MIL-STD-202, condition A or B.
At 25°C - 100,000 megohms minimum.
At 150°C - 15,000 megohms minimum.

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Capacitance: Method 305 of MIL-STD-202.
Measurement frequency:
Capacitance values of 1000 pF or less - 1 MHz ±100 kHz.
Capacitance values of greater than 1000 pF - 1 kHz ±100 Hz.

Shock, medium impact: Method 205 of MIL-STD-202, condition C (50G).

Vibration, high frequency: Method 204 of MIL-STD-202, condition B.

Moisture resistance: Test method 106 of MIL-STD-202.
Insulation resistance: Not less than 100,000 megohms.
Capacitance change: Within 1 percent for the initial value or 0.5 pF, whichever is greater.
Dissipation factor - Initial requirement.

(C) Thermal shock and immersion:
Thermal shock: Test method 107 of MIL-STD-202, condition A, except temperature for step 3 shall be 150° $^{+3°}_{-0°}$ C.
Immersion: Test method 104 of MIL-STD-202, condition A.
Insulation resistance: Not less than 100,000 megohms.
Capacitance change: Within 1 percent of the initial value or 0.5 pF, whichever is greater.

Life: 2000 ±12 hours at 150° $^{+3°}_{-0°}$ C.
Insulation resistance: Greater than 100,000 megohms.
Capacitance change: Within 1 percent of the initial value or 0.5 pF, whichever is greater.

Temperature coefficent and capacitance drift: Within the limits specified for the applicable characteristic.

Torque: In accordance with MIL-C-10950.
Mounting - normal mounting means.
Load - 2 ounce-inches.

NOTES:

1. Conversion of dash numbers to part numbers is as shown in table II.
(C) 2. Characteristic E has been replaced by characteristic F.

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TABLE II. Dash number to part number conversion.

Dash number in MIL-C-10950/8A		Converted to part number 1/ in
Characteristic D	Characteristic E	MIL-C-10950/8B and 8C
0001	0002	CB55R-680F
0003	0004	CB55R-101F
0005	0006	CB55R-151F
0007	0008	CB55R-221F
0009	0010	CB55R-331F
0011	0012	CB55R-471F
0013	0014	CB56R-681F
0015	0016	CB56R-821F
0017	0018	CB56R-102F
0019	0020	CB57R-122F
0021	0022	CB57R-152F
0023	0024	CB57R-182F
0025	0026	CB57R-222F
0027	0028	CB57R-242F
0029	0030	CB55R-680G
0031	0032	CB55R-101G
0033	0034	CB55R-151G
0035	0036	CB55R-221G
0037	0038	CB55R-331G
0039	0040	CB55R-471G
0041	0042	CB56R-681G
0043	0044	CB56R-821G
0045	0046	CB56R-102G
0047	0048	CB57R-122G
0049	0050	CB57R-152G
0051	0052	CB57R-182G
0053	0054	CB57R-222G
0055	0056	CB57R-242G
0057		CB55RD220J
0058		CB55RD330J
0059		CB55RD470J
0060	0061	CB55R-680J
0062	0063	CB55R-101J
0064	0065	CB55R-151J
0066	0067	CB55R-221J
0068	0069	CB55R-331J
0070	0071	CB55R-471J
0072	0073	CB56R-681J
0074	0075	CB56R-821J
0076	0077	CB56R-102J

1/ See footnote at end of table.

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TABLE II. Dash number to part number conversion (continued).

Dash number in MIL-C-10950/8A		Converted to part number 1/ in MIL-C-10950/8B and /8C
Characteristic D	Characteristic E	
0078	0079	CB57R-122J
0080	0081	CB57R-152J
0082	0083	CB57R-182J
0084	0085	CB57R-222J
0086	0087	CB57R-242J
0088		CB55RB050K
	0089	CB55RD100K
	0090	CB55RD150K
	0091	CB55RD220K
	0092	CB55RD330K
	0093	CB55RD470K
	0094	CB55RD680K
	0095	CB55RD101K
	0096	CB55RD151K
	0097	CB55RD221K
	0098	CB55RD331K
	0099	CB55RD471K
	0100	CB56RD681K
	0101	CB56RD821K
	0102	CB56RD102K
	0103	CB57RD122K
	0104	CB57RD152K
	0105	CB57RD182K
	0106	CB57RD222K
	0107	CB57RD242K

1/ The complete part number shall include symbols for terminal assembly and characteristic, as applicable.

Custodians:
Army - EI
Navy - EC
Air Force - 11

Review activities:
Army - AR
Navy - SH
Air Force - 17, 85
DLA - ES

User activities:
Navy - AS, CG, OS

Preparing activity:
Navy - EC

Agent:
DLA - ES

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   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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